Product List Management System Using Singly Linked List

1: Problem Statement:

To develop a Product List Management System using a singly linked list to store prroduct details and perform insertion, deletion, searching, and display operation efficiently.

2. Objective:

• To store product details using a singly linked list. • To insert new product records into the list.

• To search for a product using its ID.

• To delete a product from the list.

• To display all product details.

3. Algorithm:

1. Start.

2. Initialize the head pointer to NULL.

3. Display the menu: Insert Product, Display Products, Search Product, Delete Product, and Exit.

4. Accept the user's choice.

5. Perform the selected operation:

6. • Insert: Accept product ID, name, and price. Create a new node and add it to the end of the linked list.

7. • Display: Traverse the linked list and display all product details.

8. • Search: Accept a product ID and search for the product in the linked list.

9. • Delete: Accept a product ID, find the product, and delete its node. If it is not found, display an appropriate message.

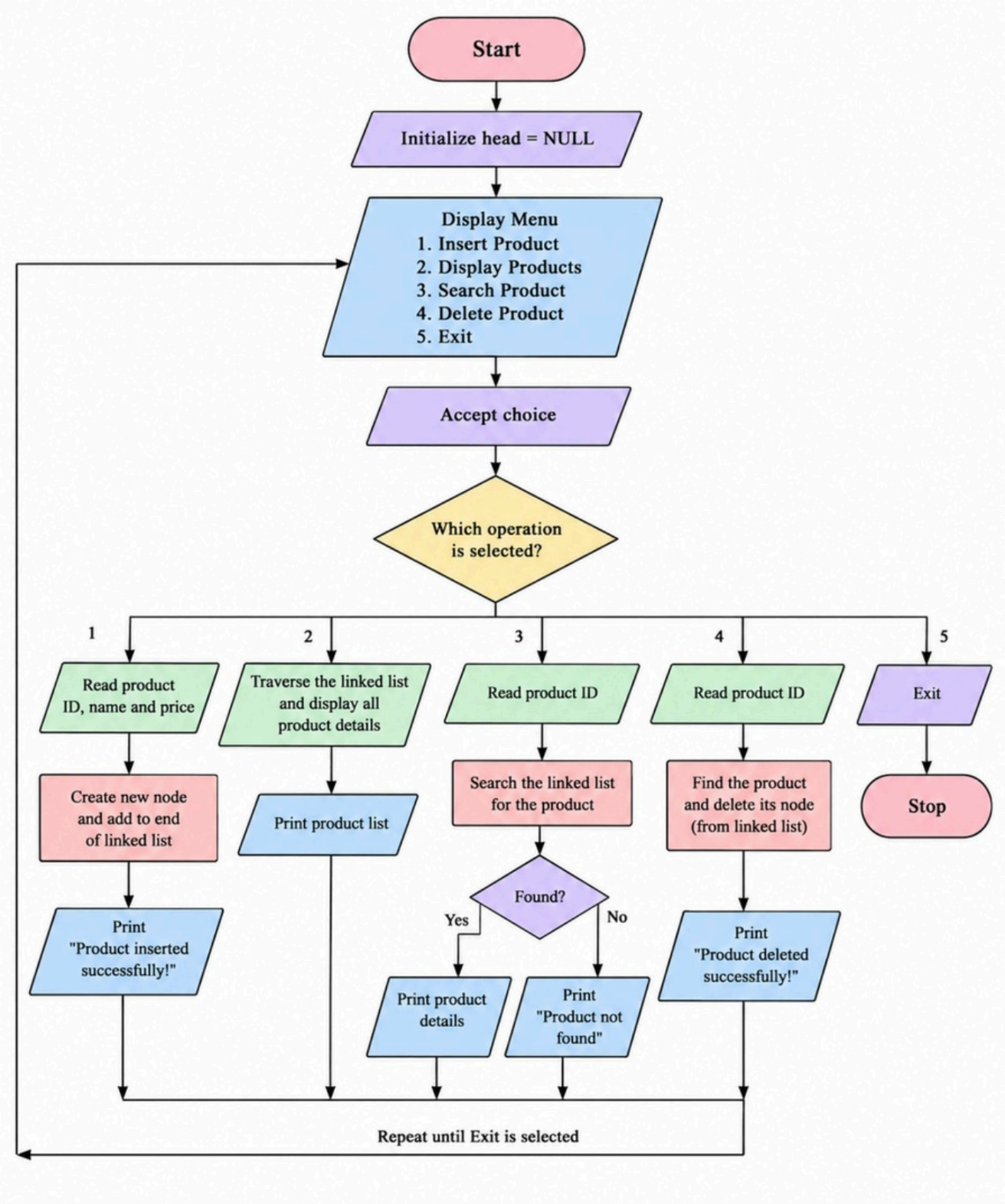
10. • Exit: Terminate the program.

11. Repeat the menu until the user selects Exit.

12. Stop.

Flowchart:

The flowchart below represents the menu-driven operations of the system.



5. Conclusion

The Product List Management System was designed using a singly linked list to manage product details. It supports insertion, deletion, searching, and display operations and helps demonstrate the use of linked lists and dynamic memory allocation in C++.